

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Winter Examination – 2024**

**Course: B. Tech**

**Branch : Mechanical Engineering**

**Semester :IV**

**Subject Code & Name: BTMEC403\_Y19 Strength of Materials**

**Max Marks: 60**

**Date:24/12/2024**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

|             |                                                                                                                           | (Level/CO) | Marks     |
|-------------|---------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1</b> | <b>Objective type questions. (Compulsory Question)</b>                                                                    |            | <b>12</b> |
| 1           | Which law is also called as the elasticity law?                                                                           | CO 1       | 1         |
|             | a. Poisson's law      b. Bernoulli's      c. Stress law      d. Hooke's law                                               |            |           |
| 2           | What is the formula for calculating stress?                                                                               | CO 1       | 1         |
|             | a. Stress = Area / Force      b. Stress = Mass × Acceleration      c. Stress = Force / Area      d. Stress = Force × Area |            |           |
| 3           | The property by which a body returns to its original shape after removal of the force is called _____                     | CO2        | 1         |
|             | a. a) Plasticity      b. Elasticity      c. Ductility      d. Ductility                                                   |            |           |
| 4           | The materials which have the same elastic properties in all directions are called..                                       | CO2        | 1         |
|             | a. Isotropic      b. Brittle      c. Homogeneous      d. Hard                                                             |            |           |
| 5           | Which stress comes when there is an eccentric load applied?                                                               | CO1        | 1         |
|             | a. Shear stress      b. Bending stress      c. Tensile stress      d. Thermal stress                                      |            |           |
| 6           | What is the expression of the bending equation?                                                                           | CO2        | 1         |
|             | a. $M/I = \sigma/y = E/R$ b. $M/R = \sigma/y = E/I$ c. $M/y = \sigma/R = E/I$ d. $M/I = \sigma/R = E/y$                   |            |           |
| 7           | What is the formula for shear stress?                                                                                     | CO1        | 1         |
|             | a. Shear resistance / shear area      b. Force/unit area      c. Bending strain/area      d. Shear stress/length          |            |           |
| 8           | The unit of Young's modulus is                                                                                            | CO1        | 1         |
|             | a. mm/mm      b. kg/cm      c. kg      d. $\text{kg/cm}^2$                                                                |            |           |
| 9           | The ultimate tensile stress of mild steel compared to ultimate compressive stress is                                      | CO2        | 1         |
|             | a. same      b. more      c. less      d. unpredictable                                                                   |            |           |
| 10          | The neutral axis is of the cross-section a % beam is that axis at which the bending stress is                             | CO2        | 1         |

|             |                                                                                                                                                                                                                                                                                                                                                                             |                       |                              |                 |     |           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------|-----------------|-----|-----------|
|             | a. zero                                                                                                                                                                                                                                                                                                                                                                     | b. minimum            | c. maximum                   | d. infinity     |     |           |
| 11          | Euler's formula holds good for                                                                                                                                                                                                                                                                                                                                              |                       |                              |                 | CO1 | 1         |
|             | a. short columns                                                                                                                                                                                                                                                                                                                                                            | b. long columns       | c. both short & long columns | d. weak columns |     |           |
| 12          | The deformation per unit length is called                                                                                                                                                                                                                                                                                                                                   |                       |                              |                 | CO1 | 1         |
|             | a. tensile stress                                                                                                                                                                                                                                                                                                                                                           | b. compressive stress | c. shear stress              | d. strain       |     |           |
|             |                                                                                                                                                                                                                                                                                                                                                                             |                       |                              |                 |     |           |
| <b>Q. 2</b> | <b>Solve the following.</b>                                                                                                                                                                                                                                                                                                                                                 |                       |                              |                 |     | <b>12</b> |
| A)          | Define the following terms:<br>i) Poisson's Ratio    2) Modulus of Elasticity    3) Hook's Law                                                                                                                                                                                                                                                                              |                       |                              |                 | CO1 | 6         |
| B)          | At a point in a strained material, direct stress of 102 MPa (tensile) and 76.5 MPa (compressive) exist on two perpendicular planes. These planes also carry shear stresses. The major principal stress is 127.5 MPa. Find shear stresses on these planes. Also find maximum shear stress.<br><br>Data: $\sigma_x = 102$ MPA, $\sigma_y = -76.5$ Mpa, $\sigma_1 = 127.5$ MPa |                       |                              |                 | CO3 | 6         |
|             |                                                                                                                                                                                                                                                                                                                                                                             |                       |                              |                 |     |           |
| <b>Q.3</b>  | <b>Solve the following.</b>                                                                                                                                                                                                                                                                                                                                                 |                       |                              |                 |     | <b>12</b> |
| A)          | Explain middle third rule and kernel of a section.                                                                                                                                                                                                                                                                                                                          |                       |                              |                 | CO1 | 6         |
| B)          | A lift weighting 20kN is to function at a speed of 1 m/s. Length of rope connecting lift is 40 m. If yield stress for rope material is 300 MPa, find suitable diameter of rope assuming factor of safety 2. Assume $E = 200$ GPa.                                                                                                                                           |                       |                              |                 | CO2 | 6         |
|             |                                                                                                                                                                                                                                                                                                                                                                             |                       |                              |                 |     |           |
| <b>Q. 4</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                      |                       |                              |                 |     | <b>12</b> |
| A)          | Derive the flexural formula.                                                                                                                                                                                                                                                                                                                                                |                       |                              |                 | CO2 | 6         |
| B)          | A square column of 100 mm x 100 mm cross section has a concentric longitudinal hole of 50 mm diameter. The length of the column is 5 m. one end fixed and other hinged. Determine the Euler's buckling load assuming $E = 200$ Gpa.                                                                                                                                         |                       |                              |                 | CO3 | 6         |
| C)          | Power transmitted by a shaft having 60 mm diameter rotating at 150 r.p.m, if maximum permissible shear stress = 80 Mpa                                                                                                                                                                                                                                                      |                       |                              |                 | CO3 | 6         |
|             |                                                                                                                                                                                                                                                                                                                                                                             |                       |                              |                 |     |           |
| <b>Q.5</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                      |                       |                              |                 |     | <b>12</b> |
| A)          | Derive the equation of torsion for circular shaft.                                                                                                                                                                                                                                                                                                                          |                       |                              |                 | CO2 | 6         |
|             |                                                                                                                                                                                                                                                                                                                                                                             |                       |                              |                 |     |           |

|      |                                                                                                                                                                                                                                                                                                                                           |    |  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--|
|      |                                                                                                                                                                                                                                                                                                                                           |    |  |
| B)   | <p>A simply supported beam ABC is loaded as shown in figure. Determine the location at which concentrated load 5 kN must act from end A to make the reactions at A and B equal. Draw SFD and BMD.</p>                                                                                                                                     | 6  |  |
| C)   | <p>A Steel shaft of solid circular cross section has to transmit 200 kW at 180 r.p.m. The maximum shear stress is not to exceed 50 MPa and the angle of twist must not be more than 1.60 in a length of 2.6 m. Design suitable diameter of shaft. Take <math>G=80</math> GPa.</p>                                                         | 6  |  |
| Q. 6 | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                           | 12 |  |
| A)   | Derive the equation of shear force, bending moment and intensity of loading.                                                                                                                                                                                                                                                              | 6  |  |
| B)   | Derive the shear stress formula.                                                                                                                                                                                                                                                                                                          | 6  |  |
| C)   | <p>A beam is fixed at both ends is loaded transversely by total uniformly distributed load of 32 kN. It is found that deflection at centre is <math>\frac{1}{11325}</math> of span. Now if this transverse load is removed and beam is loaded axially, find out safe axial load for the condition such that it fails by buckling only</p> | 6  |  |
|      | *** End ***                                                                                                                                                                                                                                                                                                                               |    |  |